

Lab: Production-Grade 3-Tier Secure VPC Architecture

Objective: Build a complete, secure, highly available VPC from scratch with proper public/private isolation, routing, security layers, and basic connectivity.

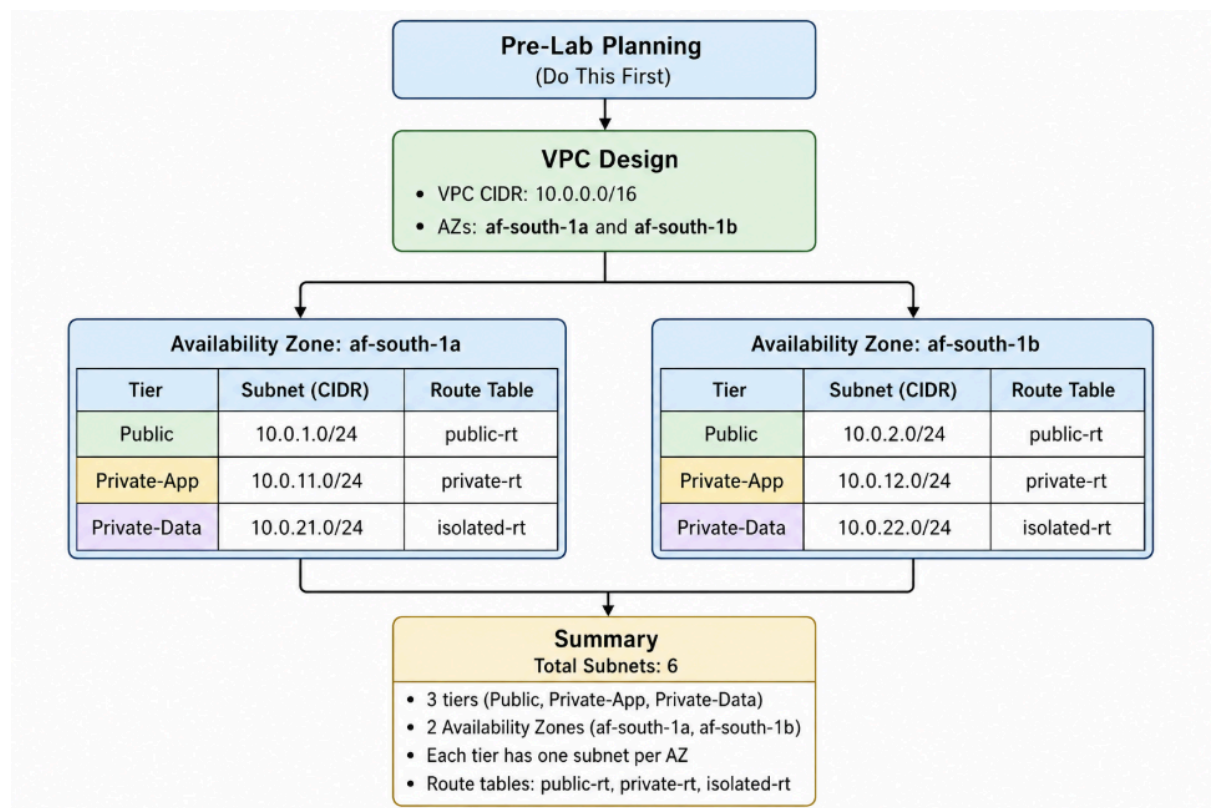
Skills Covered:

- VPC Creation & CIDR Planning
- Public vs Private Subnets
- Route Tables, IGW & NAT Gateway
- Security Groups + NACLs
- Basic understanding of connectivity options

Region: af-south-1 (Cape Town)

Pre-Lab Planning (Do This First)

VPC Design:



- VPC CIDR: 10.0.0.0/16
- AZs: **af-south-1a** and **af-south-1b**

Subnets (6 total):

Tier	AZ-1a CIDR	AZ-1b CIDR	Route Table
Public	10.0.1.0/24	10.0.2.0/24	public-rt
Private-App	10.0.11.0/24	10.0.12.0/24	private-rt
Private-Data	10.0.21.0/24	10.0.22.0/24	isolated-rt

Phase 1: Create the VPC Foundation (Days 11–12)

1. Go to **VPC** → **Create VPC**

- Name: week3-capstone-vpc
- IPv4 CIDR: 10.0.0.0/16
- Create

vpc-0f224e8e25bd7fcaa / week3-capstone-vpc
Actions

Details Info

VPC ID
vpc-0f224e8e25bd7fcaa

DNS resolution
Enabled

Main network ACL
acl-02d624ea9c36ae46a

IPv6 CIDR (Network border group)
–

Encryption control ID
–

State
Available

Tenancy
default

Default VPC
No

Network Address Usage metrics
Disabled

Encryption control mode
–

Block Public Access
Off

DHCP option set
–

IPv4 CIDR
10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups
–

DNS hostnames
Disabled

Main route table
rtb-0157edc9a3aba3027

IPv6 pool
–

Owner ID
–

Resource map CIDRs Flow logs Tags Integrations

Resource map Info

VPC
Your AWS virtual network
week3-capstone-vpc

Subnets (0)
Subnets within this VPC

Route tables (1)
Route network traffic to resources
rtb-0157edc9a3aba3027

2. Create 6 Subnets (follow the table above)

- Name them clearly: public-az1a, private-app-az1a, private-data-az1a, public-az1b, private-app-az1b, private-data-az1b.

The screenshot shows the AWS VPC console with a notification at the top: "You have successfully created 6 subnets: subnet-0c4a4f7a270b567cf, subnet-09ac2a5f376bf4516, subnet-038df7d33485d62ed, subnet-01bf8d29d5f31ac79, subnet-0ac9fc0501242d7db, subnet-0e6198e8933a2c32b". Below the notification, a table lists the subnets:

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
private-app-az1b	subnet-01bf8d29d5f31ac79	Available	vpc-0f224e8e25bd7fcaa week...	Off	10.0.12.0/24
private-data-az1a	subnet-0ac9fc0501242d7db	Available	vpc-0f224e8e25bd7fcaa week...	Off	10.0.21.0/24
private-data-az1b	subnet-0e6198e8933a2c32b	Available	vpc-0f224e8e25bd7fcaa week...	Off	10.0.22.0/24
private-app-az1a	subnet-038df7d33485d62ed	Available	vpc-0f224e8e25bd7fcaa week...	Off	10.0.11.0/24
public-az1a	subnet-0c4a4f7a270b567cf	Available	vpc-0f224e8e25bd7fcaa week...	Off	10.0.1.0/24
public-az1b	subnet-09ac2a5f376bf4516	Available	vpc-0f224e8e25bd7fcaa week...	Off	10.0.2.0/24

The screenshot shows the details for subnet-0c4a4f7a270b567cf / public-az1a. The details are as follows:

Details	Subnet ID	Subnet ARN	State	Block Public Access
Subnet ID	subnet-0c4a4f7a270b567cf	arn:aws:ec2:af-south-1:subnets/subnet-0c4a4f7a270b567cf	Available	Off
IPv4 CIDR	10.0.1.0/24	Available IPv4 addresses	IPv6 CIDR	IPv6 CIDR association ID
Availability Zone	afs1-az1 (af-south-1a)	251		
Network ACL	-	Network border group	VPC	Route table
Auto-assign customer-owned IPv4 address	No	af-south-1	vpc-0f224e8e25bd7fcaa week3-capstone-vpc	-
IPv6 CIDR reservations	-	Default subnet	Auto-assign public IPv4 address	Auto-assign IPv6 address
Resource name DNS AAAA record	Disabled	No	Yes	No
		Customer-owned IPv4 pool	Outpost ID	IPv4 CIDR reservations
		-	Hostname type	-
		IPv6-only	Owner	Resource name DNS A record
		No		Disabled
		DNS64		
		Disabled		

The screenshot shows the details for subnet-038df7d33485d62ed / private-app-az1a. The details are as follows:

Details	Subnet ID	Subnet ARN	State	Block Public Access
Subnet ID	subnet-038df7d33485d62ed	arn:aws:ec2:af-south-1:subnets/subnet-038df7d33485d62ed	Available	Off
IPv4 CIDR	10.0.11.0/24	Available IPv4 addresses	IPv6 CIDR	IPv6 CIDR association ID
Availability Zone	afs1-az1 (af-south-1a)	251		
Network ACL	-	Network border group	VPC	Route table
Auto-assign customer-owned IPv4 address	No	af-south-1	vpc-0f224e8e25bd7fcaa week3-capstone-vpc	-
IPv6 CIDR reservations	-	Default subnet	Auto-assign public IPv4 address	Auto-assign IPv6 address
Resource name DNS AAAA record	Disabled	No	No	No
		Customer-owned IPv4 pool	Outpost ID	IPv4 CIDR reservations
		-	Hostname type	-
		IPv6-only	Owner	Resource name DNS A record
		No		Disabled
		DNS64		
		Disabled		

Details

Subnet ID
subnet-0ac9fc0501242d7dbIPv4 CIDR
10.0.21.0/24Availability Zone
af-south-1aNetwork ACL
-Auto-assign customer-owned IPv4 address
NoIPv6 CIDR reservations
-Resource name DNS AAAA record
DisabledSubnet ARN
arn:aws:ec2:af-south-1:123456789012:subnet/subnet-0ac9fc0501242d7dbAvailable IPv4 addresses
251Network border group
af-south-1Default subnet
NoCustomer-owned IPv4 pool
-IPv6-only
NoDNS64
DisabledState
AvailableIPv6 CIDR
-VPC
vpc-0f224e8e25bd7fcaa | week3-capstone-vpcAuto-assign public IPv4 address
NoOutpost ID
-Hostname type
IP nameOwner
123456789012Block Public Access
OffIPv6 CIDR association ID
-Route table
-Auto-assign IPv6 address
NoIPv4 CIDR reservations
-Resource name DNS A record
Disabled

Phase 2: Routing & Internet Access

3. Create Internet Gateway

- Name: week3-igw
- Attach to your VPC

VPC > Internet gateways > igw-0bf3f74d0d159abee

Internet gateway igw-0bf3f74d0d159abee successfully attached to vpc-0f224e8e25bd7fcaa

igw-0bf3f74d0d159abee / week3-igw

Details

Internet gateway ID
igw-0bf3f74d0d159abee

State
Attached

VPC ID
vpc-0f224e8e25bd7fcaa | week3-capstone-vpc

Owner
123456789012

Tags (1)

Search tags

Key	Value
Name	week3-igw

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

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4. Create 3 Route Tables

☐	public-rt	rtb-01a163aa79b07184c	2 subnets	-	No	vpc-0f224e8e25bd7fcaa week3-capstone-vpc
☐	private-rt	rtb-06b828c9381f4436c	2 subnets	-	No	vpc-0f224e8e25bd7fcaa week3-capstone-vpc
☐	isolated-rt	rtb-0870a7a44ad25b145	2 subnets	-	No	vpc-0f224e8e25bd7fcaa week3-capstone-vpc

- public-rt → Add route 0.0.0.0/0 → week3-igw

The screenshot shows the AWS VPC console for a route table named 'rtb-01a163aa79b07184c / public-rt'. The left sidebar shows the 'VPC dashboard' with a 'Filter by VPC' dropdown and a list of 'Virtual private cloud' resources. The main content area shows the 'Details' tab for the route table. It includes fields for 'Route table ID' (rtb-01a163aa79b07184c), 'VPC' (vpc-0f224e8e25bd7fcaa | week3-capstone-vpc), 'Main' (No), 'Owner ID' (redacted), 'Explicit subnet associations' (2 subnets), and 'Edge associations' (-). Below this is a tabbed interface with 'Routes' selected. The 'Routes' section shows a table with 2 routes:

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0bf3f74d0d159abee	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

- private-rt → (we'll add NAT later)

The screenshot shows the AWS VPC console for a route table named 'rtb-06b828c9381f4436c / private-rt'. The left sidebar shows the 'VPC dashboard' with a 'Filter by VPC' dropdown and a list of 'Virtual private cloud' resources. The main content area shows the 'Details' tab for the route table. It includes fields for 'Route table ID' (rtb-06b828c9381f4436c), 'VPC' (vpc-0f224e8e25bd7fcaa | week3-capstone-vpc), 'Main' (No), 'Owner ID' (redacted), 'Explicit subnet associations' (2 subnets), and 'Edge associations' (-). Below this is a tabbed interface with 'Routes' selected. The 'Routes' section shows a table with 1 route:

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

- isolated-rt → Only local route (no internet)

The screenshot shows the AWS VPC console for a route table named 'rtb-0870a7a44ad25b145 / isolated-rt'. The left sidebar shows the 'VPC dashboard' with a 'Filter by VPC' dropdown and a list of 'Virtual private cloud' resources. The main content area shows the 'Details' tab for the route table. It includes fields for 'Route table ID' (rtb-0870a7a44ad25b145), 'VPC' (vpc-0f224e8e25bd7fcaa | week3-capstone-vpc), 'Main' (No), 'Owner ID' (redacted), 'Explicit subnet associations' (2 subnets), and 'Edge associations' (-). Below this is a tabbed interface with 'Routes' selected. The 'Routes' section shows a table with 1 route:

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

5. Associate Subnets

- Public subnets → public-rt
- Private-App subnets → private-rt
- Private-Data subnets → isolated-rt

Available subnets (6)

Filter subnet associations

☐	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	private-app-az1b	subnet-01bf8d29d5f31ac79	10.0.12.0/24	–	rtb-06b828c9381f4436c / private-rt
☐	private-data-az1a	subnet-0ac9fc0501242d7db	10.0.21.0/24	–	rtb-0870a7a44ad25b145 / isolated-rt
☐	private-data-az1b	subnet-0e6198e8933a2c32b	10.0.22.0/24	–	rtb-0870a7a44ad25b145 / isolated-rt
☐	private-app-az1a	subnet-038df7d33485d62ed	10.0.11.0/24	–	rtb-06b828c9381f4436c / private-rt
☐	public-az1a	subnet-0c4a4f7a270b567cf	10.0.1.0/24	–	rtb-01a163aa79b07184c / public-rt
☐	public-az1b	subnet-09ac2a5f376bf4516	10.0.2.0/24	–	rtb-01a163aa79b07184c / public-rt

6. Create NAT Gateway (for Private-App outbound)

- In one **public** subnet (e.g. public-az1a)
- Allocate Elastic IP
- Name: week3-nat-az1a

VPC > NAT gateways > nat-1179d9173ce6370b2

VPC dashboard < AWS Global View [↗](#)

Filter by VPC

▼ Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways**
- Peering connections
- Route servers

▼ Security

nat-1179d9173ce6370b2 / week3-nat-az1a

Actions

Details

NAT gateway ID nat-1179d9173ce6370b2	Availability mode Regional	State Available	State message Info –
NAT gateway ARN arn:aws:ec2:af-south-1:1179d9173ce6370b2:natgateway/nat-1179d9173ce6370b2	Connectivity type Public	Created Tuesday, May 19, 2026 at 22:36:46 GMT +1	Deleted –
VPC vpc-0f224e8e25bd7fcaa / week3-capstone-vpc	Method of EIP allocation Automatic		

IP addresses | Monitoring | Flow logs | Tags

Associated IP addresses (1) [Edit IP address associations](#)

Search

IP address	Status	Availability Zone	Allocation ID	Asso
15.240.96.221	Succeeded	afs1-az2 (af-south-1b)	eipalloc-046d7fe30faf555b	eipai

- Add route in private-rt: 0.0.0.0/0 → NAT Gateway

Updated routes for rtb-06b828c9381f4436c / private-rt successfully

rtb-06b828c9381f4436c / private-rt

Details

Route table ID: rtb-06b828c9381f4436c

VPC: vpc-0f224e8e25bd7fcaa | week3-capstone-vpc

Main: No

Owner ID: [REDACTED]

Explicit subnet associations: 2 subnets

Edge associations: -

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-1179d9173ce6370b2	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Phase 3: Security Layers

7. Create Security Groups:

- sg-alb: Allow HTTP/HTTPS (80/443) from 0.0.0.0/0

sg-0d5c2b860eeb2944e - SG-alb

Details

Security group name: SG-alb

Security group ID: sg-0d5c2b860eeb2944e

Description: internet access

VPC ID: vpc-0f224e8e25bd7fcaa

Owner: [REDACTED]

Inbound rules count: 2 Permission entries

Outbound rules count: 1 Permission entry

Inbound rules (2)

Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
sgr-034ad76a84a22a06c	IPv4	HTTPS	TCP	443	0.0.0.0/0	-
sgr-00ee3d72b501010c6	IPv4	HTTP	TCP	80	0.0.0.0/0	-

- sg-app: Allow port 8080 from sg-alb

VPC dashboard < AWS Global View Filter by VPC

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Security
Network ACLs

Security group (sg-0aca6451bfb0c4d56 | SG-app) was created successfully
Details

sg-0aca6451bfb0c4d56 - SG-app Actions

Details

Security group name SG-app	Security group ID sg-0aca6451bfb0c4d56	Description access from ALB	VPC ID vpc-0f224e8e25bd7fcaa
Owner	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules Outbound rules Sharing VPC associations Related resources - new Tags

Inbound rules (1) Manage tags Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-03a438468fbad047f	-	Custom TCP	TCP	8080

- sg-rds: Allow port 3306 (or 5432) from sg-app

VPC dashboard < AWS Global View Filter by VPC

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Security group (sg-0b6fde65d30f5cdc5 | SG-rds) was created successfully
Details

sg-0b6fde65d30f5cdc5 - SG-rds Actions

Details

Security group name SG-rds	Security group ID sg-0b6fde65d30f5cdc5	Description database	VPC ID vpc-0f224e8e25bd7fcaa
Owner	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules Outbound rules Sharing VPC associations Related resources - new Tags

Inbound rules (1) Manage tags Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-037fd975385ac8c6e	-	MySQL/Aurora	TCP	3306

8. Create Custom NACLs (optional but recommended for practice):

- One for public, one for private, one for data tier
- Remember ephemeral ports (1024-65535) for NACLs

Phase 4: Deploy Resources & Test

9. Launch EC2 in Public Subnet (Bastion/Test)

- Use sg-alb or create a simple one
- Auto-assign Public IP: Enable

10. Launch EC2 in Private-App Subnet

- Attach sg-app
- No public IP

Instances (4) Info

Last updated less than a minute ago

Refresh

Connect

Instance state ▾

Actions ▾

Launch instances

⌵

Saved filter sets

Choose filter set ▾

⏪ 1 ⏩

⚙️

☐	Name ↗ ▾	Instance ID	Instance state ▾	Instance type ▾	Status check	Availability Zone ▾	Public IPv4 DNS	▾	P
☐	private-app-se...	i-08161d44f13478df9	Running 🔍 🔍	t3.micro	3/3 checks passed	af-south-1b	-		-
☐	public-server-2	i-0be86b700ce7356a2	Running 🔍 🔍	t3.micro	3/3 checks passed	af-south-1b	-		1
☐	public-server	i-06b1aa8430d0af93f	Running 🔍 🔍	t3.micro	3/3 checks passed	af-south-1a	-		1
☐	private-app-se...	i-0142932a0bdb6a8eb	Running 🔍 🔍	t3.micro	3/3 checks passed	af-south-1a	-		-

11. Test Connectivity:

- From public EC2 → Should reach internet

```
[ec2-user@ip-10-0-1-100:~]
iges1@ZSER MINGw64 ~/OneDrive/desktop/New folder
$ ssh -i "NetOps.pem" ec2-user@13.247.66.205
The authenticity of host '13.247.66.205 (13.247.66.205)' can't be established.
ED25519 key fingerprint is: SHA256:J+MigE60kSUFF9/g8c4dP4whldwq3I8K687F8SDxxu
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.247.66.205' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html

      #
     _#_#####
    /  #####\   Amazon Linux 2023
   /    ##### \
  /      ##     \
 /        #       \
/          V-----> https://aws.amazon.com/linux/amazon-linux-2023
|         /
|        /
|       /
|      /
|     /
|    /
|___/

[ec2-user@ip-10-0-1-100 ~]$ curl -I https://www.google.com
HTTP/2 200
content-type: text/html; charset=ISO-8859-1
content-security-policy-report-only: object-src 'none';base-uri 'self';script-src 'nonce-AmSunRkp13XZhVynF64opw' 'strict-dynam
ic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri https://csp.withgoogle.com/csp/gws/other-hp
accept-ch: Sec-CH-Prefers-Color-Scheme
p3p: CP="This is not a P3P policy! See g.co/p3phelp for more info."
date: Tue, 19 May 2026 22:49:57 GMT
server: gws
x-xss-protection: 0
x-frame-options: SAMEORIGIN
expires: Tue, 19 May 2026 22:49:57 GMT
cache-control: private
set-cookie: AEC=AaJmaStjg1ckGF1WqNi_8R10pFkjwHV9ZL8HwiojosRIYwm6RSC3KQOz_hc; expires=Sun, 15-Nov-2026 22:49:57 GMT; path=/; do
main=.google.com; Secure; HttpOnly; SameSite=lax
set-cookie: NID=531=R7OzfR0j1bQ5qG13uolFZaf0Ym9mtapFRtWziSLtbWN7EGT-Ce4f5rOE96Jhw5PSNhLFLEROYUN-2WXEnEm9cvkReij026Nvi3pZM6U
FLH3mAPR3SEBzr77QgoQLsgxkdYCWp9X0-EaaUCSRqmbN1-FSHMNDB9N3QP19PeV_EZWwYv_eqEX_QyyMB5j5mAEE3-2EE5W1CqnImH; expires=Wed, 18-No
v-2026 22:49:57 GMT; path=/; domain=.google.com; HttpOnly
alt-svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

[ec2-user@ip-10-0-1-100 ~]$ curl ifconfig.me # Should show a public IP
13.247.66.205[ec2-user@ip-10-0-1-100 ~]$ ping -c 3 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=29.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=29.0 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=112 time=29.0 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 29.032/29.037/29.048/0.007 ms
[ec2-user@ip-10-0-1-100 ~]$
```


Question

I have a question for whoever reads this is it impossible to ssh into the private-app-EC2?

Do tell me your answer.

Thank You